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Classification of network security attacks using novel enhanced restricted Boltzmann machine-based bitterling fish optimization algorithm

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Abstract

In the past, intrusion detection was mostly dependent on established techniques like protocol control, anti-virus software models, firewalls, and the encryption-decryption approach. Although these techniques were effective in detecting a small count of attacks, they had trouble detecting many attacks and had a significant false positive rate. But when harmful attacks get larger, new problems arise that require an all-encompassing reaction. Hence, this article plans to perform network security attack classification using novel intelligent deep learning techniques. Initially, the data is gathered from the standard benchmark datasets, such as the CICIDS 2017 and CSE-CICIDS 2018 datasets. The collected data undergoes pre-processing using the Min-Max normalization method. Next, the features are extracted from these pre-processed data using the Principal Component Analysis (PCA) method. These extracted features undergo the final classification stage using the novel Enhanced Restricted Boltzmann Machine (ERBM), in which the parameter tuning of RBM is accomplished by the nature inspired optimization algorithm referred as Bitterling Fish Optimization (BFO) algorithm with the intention of accuracy maximization as the major fitness function. This novel ERBM-BFO classifies the final output into various attacks, such as normal, DoS attack, DDoS attack, brute force attack, botnet attack, web attack, infiltration attack, and port scan attack, respectively. The outcomes suggested that the method is effective and suitable for identifying attacks in contemporary situations. The proposed ERBM-BFO method for the network security classification model in terms of accuracy is 9.35%, 6.85%, 4.47%, and 2.19% better than LSTM + AE, CNN+LSTM, MLP, and RBM, respectively. Similarly, the proposed ERBM-BFO method for the network security classification model with respect to recall is 9.44%, 6.92%, 4.51%, and 2.20% advanced than LSTM + AE, CNN+LSTM, MLP, and RBM, respectively.

Keywords Network security attack classification, Enhanced Restricted Boltzmann Machine, Bitterling fish optimization, Min-Max Normalization method, Principal component analysis

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1 Introduction

Systems as well as frameworks for Information and Communications Technology (ICT) handle a variety of sensitive user information [1]. These attacks may be human or automatic as computer frameworks are used in more and more contexts as well as applications [2]. Many companies use traditional security solutions, like anti-spam techniques, firewalls, and antivirus software, to protect themselves from network threats [3]. Regretfully, these technologies are unable to recognize novel or intricate threats. As it's crucial to identify unusual access to secured internal networks, intrusion detection has long been a major topic of research in network security [4].

In the past, intrusion detection was mostly dependent on established techniques like anti-virus software models, protocol control, firewalls, and the encryption-decryption approach [5]. These techniques produced a high percentage of false positives and had trouble detecting a vast count of attacks, even if they were successful in recognizing a small count of attacks [6]. In particular, hackers carry out a great deal of DoS attacks, which are challenging to recognize and counter with conventional techniques [7]. Incorporating ML approaches has been the main focus of present research. They often result in higher identification rates than existing methods while reducing the overhead related to handling large-scale attacks [8]. However, traditional ML methods are severely limited in their ability to detect attacks using highly integrated characteristics [9]. Furthermore, these methods exhibit subpar performance in handling noisy traffic and multi-dimensional data [10].

In recent years, intrusion detection has been enhanced using Deep Learning (DL) algorithms. Improved learning behaviors as well as higher intrusion detection accuracy rates have been demonstrated by methods for intrusion detection on the basis of nonlinear structured deep learning frameworks [11]. Furthermore, in order to support the considerable security advancements of the DL methods, hardware designs have been improved in recent years. Network system security must be guaranteed because of how important this problem is [12]. Utilizing technologies to support management as well as security activities is essential. To make it easier to identify unusual activity as well as put safeguards in place to lessen the effects of hostile actors, these devices also need to be automated [13]. DL effectively beats traditional algorithms in just a small number of intrusion detection research papers [14–17]. The general network security threats model is diagrammatically shown in Fig. 1.

The paper contribution is as below.

- To perform the network security attack classification using novel intelligent deep learning techniques.
- To gather the data from the standard benchmark dataset such as CICIDS2017 and CSE-CICIDS2018 datasets and to pre-process it using the Min-Max normalization method.
- To extract the features from these pre-processed data using the PCA method.
- To undergo the final classification stage using the novel ERBM, in which the parameter tuning of RBM is accomplished by the nature inspired optimization algorithm referred as BFO algorithm with the intention of accuracy maximization as the major fitness function.

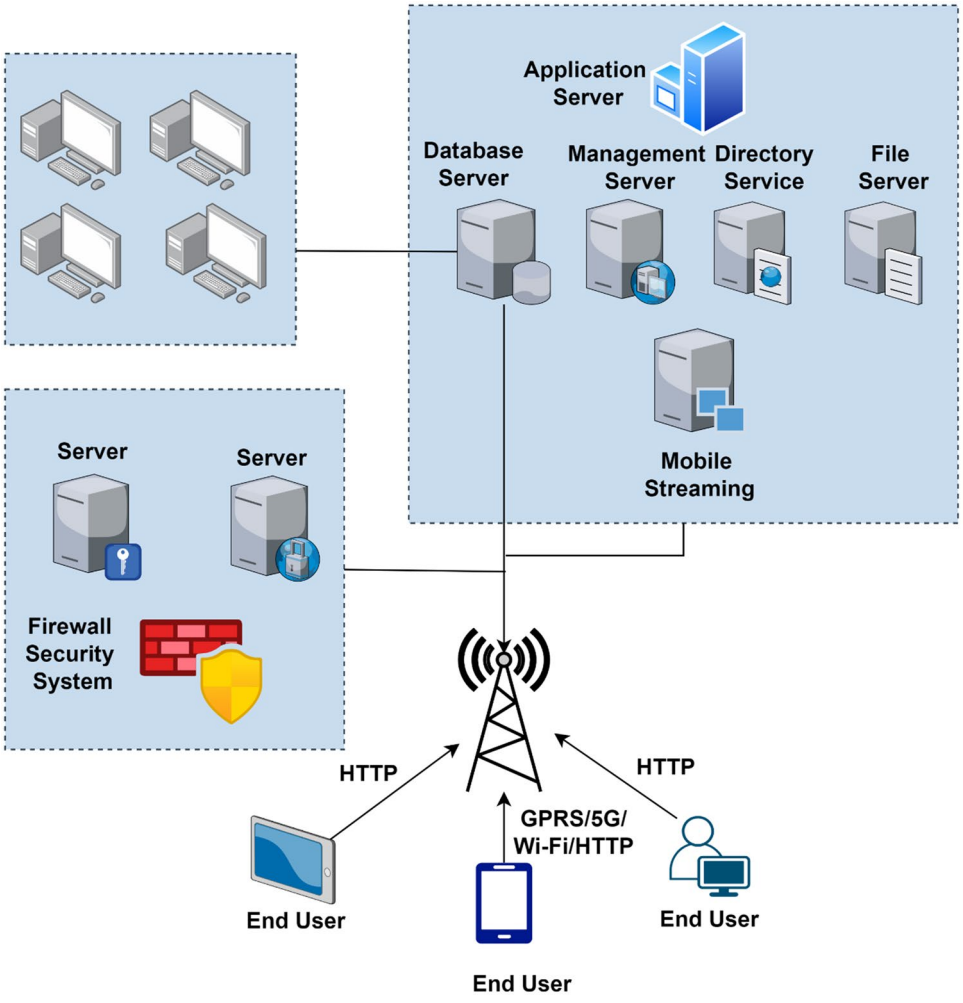


Fig. 1 General network security threats model

- To classify the final output into various attacks such as normal, DoS attack, DDoS attack, Brute Force attack, Botnet attack, Web attack, Infiltration attack, and Port Scan attack respectively by the novel ERBM-BFO.

The paper organization is as follows. Section 1 is the introduction of the network security threats classification. Section 2 is literature survey. Section 3 is proposed methodology with data collection, pre-processing using Min-Max normalization method, feature extraction using PCA, classification using ERBM, and BFO. Section 4 is results. Section 5 is conclusion.

1.1 Research motivation

Even though there exists a number of ML as well as DL methods for classifying DDoS attacks in the literature, more work has to be done to improve classification effectiveness. Due to the method's ongoing deepening, DL approaches parameter counts also rise quickly, leading to overfitting of the method. In general, to get effective results, hyperparameters are crucial. Metaheuristic methods are useful for hyperparameter tuning because the trial-and-error approach defines a time-consuming as well as inaccurate procedure.

Existing RBM-based intrusion detection methods commonly employ conventional contrastive-divergence learning or manually selected parameters, making their performance sensitive to initialization and potentially leading to slow convergence or suboptimal solutions. Likewise, recent CNN-, LSTM-, and autoencoder-based approaches can capture complex traffic patterns but often involve substantial computational overhead, extensive hyperparameter tuning, and limited interpretability in high-dimensional feature spaces. The proposed ERBM–BFO framework differs technically by combining PCA-based compact feature representation with an enhanced RBM whose influential learning parameters are adaptively optimized through the Bitterling Fish Optimization algorithm. BFO balances global exploration through oyster-search and escape behaviors with local exploitation around promising candidate solutions, thereby reducing dependence on trial-and-error parameter selection. Moreover, the framework supports multiclass identification of normal traffic and seven attack categories and is evaluated across both CICIDS2017 and CSE-CICIDS2018. Thus, the principal novelty lies not in the individual components, but in their coordinated optimization-oriented integration for improved convergence, dimensional efficiency, classification reliability, and cross-dataset applicability.

2 Related work

In 5G networks, this research developed the Modified Equilibrium Optimization Algorithm with Deep Learning-based DDoS Attack Classification (MEOADL-ADC) technique [18]. The approach fundamentally used a feature selection strategy on the basis of MEOA. The LSTM method was then used by the MEOADL-ADC approach to classify attacks. Lastly, the Tunicate Swarm Algorithm (TSA) was used to modify the LSTM method's hyperparameter. The uniqueness related to the work was demonstrated by the TSA-oriented hyperparameter as well as MEOA-oriented feature selection tweaking designs.

In the absence of attacks, a novel framework on the basis of an ensemble classifier was described that integrated the great effectiveness of traditional 2-Class classification (referred as 2C) with the improved security of 1-Class classification (referred as 1C) [19]. In the studies, eight potential adversarial attacks were examined in different circumstances. These attacks were carried out independently against the cmb and 2C-classifier architectures.

The amplitude-shift attack and replay attack are two types of attacks designed [20]. Using these two methods, two datasets were considered that comprise correlated time series information and turn them into attack datasets. A neural network method was created on the basis of LSTM to identify amplitude-shift and replay attacks.

For the purpose of performing modulation classification, a Generative Adversarial Network (GAN)-based Siamese neuron network (GSNN) was developed [21]. The developed GSNN's defence effectiveness was validated using public and simulated datasets. Simulation findings showed that the suggested approach outperformed the widely used SoftMax classifier.

In order to identify DoS attacks in WSNs, this work suggests Decision Tree (DT) algorithm with the Gini feature selection technique [22]. With respect to processing time, the suggested method accomplished much better than XGBoost, FR, and KNN classifiers, taking just 13%, 9.7%, and 2% of the time, correspondingly.

The research combined the Support Vector Machine (SVM) with the flash search method to create a fusion power network security risk evaluation method [23]. Next, using the fusion method as a base, this technique was utilized to construct an enhanced power network security risk evaluation method. The research's enhanced algorithm has a higher accuracy—96.2%—than the remaining comparative algorithms.

Seven typical classification algorithms were chosen for this research in order to detect common threats, and a comparison analysis was used to determine which detection approach was most appropriate as well as advantageous [24]. An ideal Convolutional Neural Network (CNN) identification technique that makes usage of data field features as well as classifier selection was presented and evaluated. Furthermore, the notion related to the Hamming distance was presented, which allows an improved CNN algorithm to be proposed that accomplishes reliable identification of difficult-to-identify anomalous information.

This paper suggests a 5G-oriented framework that detects and categorizes cyberattacks using the Deep Learning- Stacked LSTM (DL-SkLSTM) oriented architecture [25]. To distinguish between distinct cyberattacks, SkLSTM was employed. After carrying out an extensive examination and contrast, it was ascertained that the suggested framework surpasses many cutting-edge ML and DL methodologies.

Insider attack detection was shown to be most accurately predicted by a ML method [26]. In ML, the most practical classifier was often chosen following more assessment trials of potential methods, which might introduce bias into the methods by allowing unseen information (test data set) to seep in. As a result, overfitting happens when methods were trained often and hyperparameters are adjusted; as a result, the methods accomplished better on the training set but poorly generalize to new information. This research uses the hyperparameter adjustment and validation data set to avoid the aforementioned problems and select the optimal method from the candidate methods.

The use of Krill Based Deep Neural Network Stacked Auto Encoders (KDNN-SAE) to recognize cyber security threats in the IoT was developed [27]. Prior to dividing the dataset into the preparation as well as test segments, the suggested method initially used the information pre-processing. Flow-oriented characteristics were then taken out of the pre-processed data at that moment. By then, the Genetic Algorithm (GA) is used to choose the qualities that the algorithms would use for attribute determination. The final step in the process is to run the ML method KDNN-SAE. The accuracy related to the experimental result that was suggested was 99.37%.

This paper introduces a fusion-based multilayered SDN intrusion detection and mitigation framework called Layer-Based-InSDN Dataset-Intrusion Detection System (LB-ISD-IDS) [32]. Initially, data preprocessing is done on ISD using label encoding, Z-Score normalization, Recursive Feature Elimination (RFE), and SHapley Additive exPlanations (SHAP). ISD class imbalance is addressed by applying the K-means Synthetic Minority Over-sampling Technique (SMOTE) balancing strategy, which avoids noise amplification. The hybrid ISD attack classifier employs Temporal Convolutional Network (TCN) to extract sequential ISD attack patterns, a LightGradient Boosting Machine (LightGBM) for decision-tree-based rapid and interpretable ISD attack classification, and TabNet for selective attention-based ISD feature transformation.

This research focuses on prompt detection of pathogenic threats to paddy leaves through image analysis [33]. The inputs are pre-processed using an Adaptive Gabor

Filter (AGF), for improving the quality and reducing the noise from the image. Subsequently, using a Histogram of Oriented Gradient (HOG), the extraction of relevant disease-related information is assured. For the classification stage, an OptimizedCapsule Networks (CapsNet) is employed for identifying diseases affecting paddy foliage with high accuracy. The proposed CapsNet achieves improved generalization and robustness by integrating the Glow Worm Swarm Algorithm (GWSA) for parameter optimization. In addition to disease prediction, the proposed system integrates a Fertilizer-Based Disease Management (FBDM) approach, which suggests precise fertilization strategies based on the identified disease.

This paper proposes a Hidden Cross Shannon Entropy Markov Model (HCSEMM)-centric transition state analysis [34]. Primarily, the Papanicolaou (Pap) smear images are gathered and pre-processed. Afterward, the stain is deconvoluted by utilizing the Non-negative matrix factorization Macenko Stain Deconvolution (NMSD) approach. Then, the image is balanced by using the Chain Normalization-based Generative Adversarial Network (CNGAN). Subsequently, the cytoplasm and nucleus are segmented utilizing the Atrous Spatial Pyramid Pooling-centric U-Network (ASPPU-Net), followed by feature extraction. Thereafter, the morphological features are examined using the Coiflet Scaling Function-based Discrete Wavelet Transform (CSF-DWT). Next, CC is detected by employing the Deep Elite Pool Beluga SoftReMish-based Transfer Convolutional Neural Network (DEPBS-TCNN).

3 Problem definition and scope

The present study addresses multiclass network intrusion detection rather than dedicated insider-threat or data-leakage detection. CICIDS2017 and CSE-CICIDS2018 primarily contain labelled network-flow records representing Normal, DoS, DDoS, Brute Force, Botnet, Web Attack, Infiltration, and Port Scan activities. Although their infiltration scenarios include malicious actions originating from compromised internal hosts, they do not provide the user identities, access privileges, behavioral histories, organizational roles, or endpoint audit trails required to distinguish malicious insiders from compromised users. Accordingly, the proposed PCA–ERBM–BFO framework identifies suspicious traffic patterns observable at the network-flow level, including internally initiated infiltration behavior, but does not attribute an attack to a particular insider. References to general insider-threat and data-leakage detection have therefore been revised to avoid overstating the framework's scope. Developing a dedicated insider-threat model would require datasets such as CERT containing user activities, authentication events, file access, email, web, and endpoint logs, together with temporal behavioral profiling and identity-aware analysis. Such multimodal validation is considered an important direction for future research.

4 Proposed methodology

4.1 Proposed model

The proposed network security classification model includes various phases such as data collection, pre-processing, feature extraction, and classification. First, the CICIDS2017 and CSE-CICIDS2018 datasets, which are common benchmark datasets, are used to collect the data. The Min-Max normalization approach is used for pre-processing of the gathered data. Next, the PCA approach is used to extract the features from this

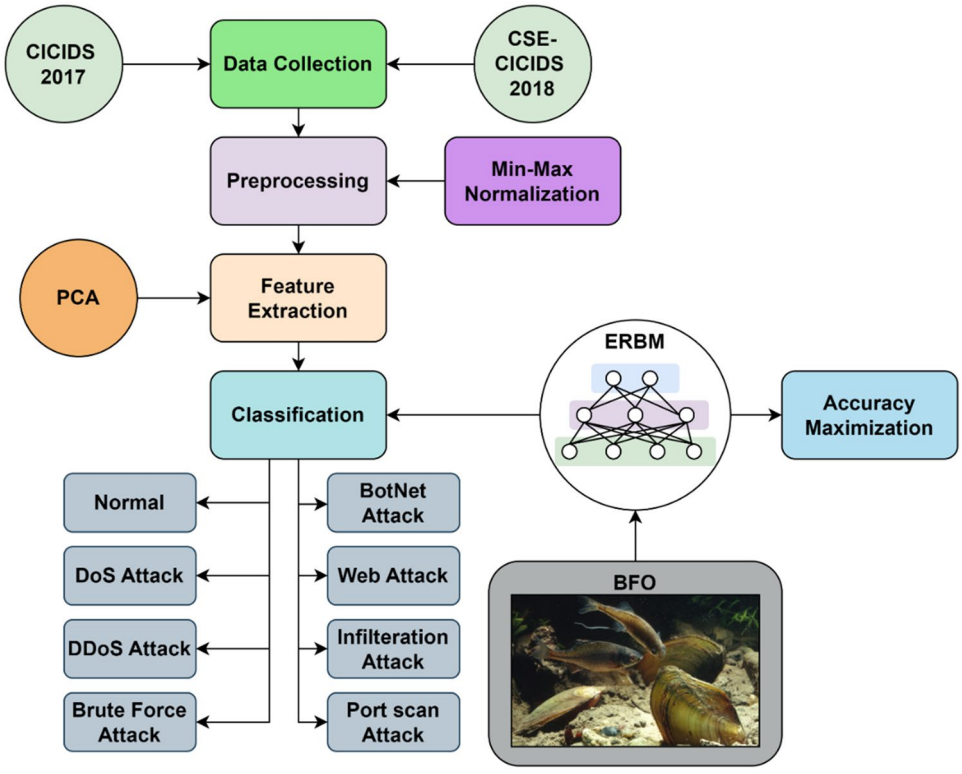


Fig. 2 Proposed network security classification model

pre-processed data. These extracted features proceed through the last stage of classification utilizing the innovative ERBM, where accuracy maximization is the primary fitness function and the parameter tuning of RBM is achieved by the nature-inspired optimization technique known as the BFO algorithm. The final output is categorized into several attacks by this innovative ERBM-BFO, including normal, DoS, DDoS, Brute Force, Botnet, Web, Infiltration, and Port Scan attacks. Figure 2 depicts the proposed network security classification model.

4.1.1 Data collection

The data for the proposed network security classification model is gathered from the standard benchmark datasets like CICIDS2017 and CSE-CICIDS2018 datasets.

4.1.1.1 CICIDS2017 dataset The goal of this dataset was to fill the gap in real-time network traffic datasets. The major recent as well as pertinent information for evaluating security frameworks is included in CICIDS-2017. But the main justification for utilizing this dataset is since it provides information on class disparities. There are more than 80 features in CICIDS-2017. Consequently, it is shown that the CICIDS-dataset has superior data dimension.

4.1.1.2 CSE-CICIDS2018 dataset The Canadian Institute for Cybersecurity (CIC) as well as the Communications Security Establishment (CSE) worked together to create this dataset by using the notion of profiles to create an extensive cybersecurity dataset. Additionally, it has abstract distribution methods for different network components, includ-

ing applications, protocols, and lower-level parts. Thirty servers and four hundred PCs make up the five departments of the target group, whereas the attacking network has fifty machines. This dataset includes 80 network traffic characteristics from intercepted network traffic in addition to log files and the network traffic of every victim machine.

4.1.2 Pre-processing using Min-Max Normalization method

The pre-processing of the gathered data for the proposed network security classification model is accomplished by the Min-Max Normalization method. Data normalization might improve the accuracy associated with the method and speed up its solution. Prior to training, the information has to be standardized and modified for best outcomes. The datasets contain outlier, duplicated, and null values. Likewise, not entire dataset samples fall into the similar range, hence normalizing them before training is necessary for accurate prediction. Redundant, null, as well as outlier values are eliminated from the datasets. Since the standard transform centres and scales every feature independently and covers all practical redundant as well as null data removal approaches, it is utilized for data normalization.

When using the Min-Max normalization approach, it is assumed that the dataset contains the characteristics $Y_1, Y_2, \cdots, Y_o$. The below equation represents the normalization of a specific data point Y_j in the group:

$$Y_j' = \frac{Y_j - Y_{Min}}{Y_{Max} - Y_{Min}} \tag{1}$$

In this case, Y_j' represents the normalized data, while Y_{Min} and Y_{Max} stand for the dataset's lowest as well as highest values, respectively.

4.1.3 Feature extraction using PCA

The features for the proposed network security classification model are extracted using the PCA approach. PCA is a potent dimensionality reduction method that preserves the major crucial data. The following equation serves as the foundation for PCA:

$$D = \frac{1}{O} Y * Y^T \tag{2}$$

In this case, Y^T describes the transpose of Y, and D describes the covariance matrix related to the standardized information Y, which consists of O samples. The relationships among the characteristics in the information are captured by matrix D.

The transformation matrix that projects the original information onto the initial l main components is represented by X_l, a crucial PCA component. Every X_l column represents a main component.

$$X_l = [W_1, W_2, W_3, \cdots, W_l] \tag{3}$$

In this case, the eigenvectors $W_1, W_2, W_3, \cdots, W_l$ represents the eigenvalues associated with the top l of the covariance matrix. X_l, in which every column represents a principal component, describes a common representation of the matrix that accomplishes the projection of the information onto the initial l principal components. The below describes the projection:

$$Y_{Proj} = Y * X_l \tag{4}$$

In this case, X_l describes the projection matrix's transpose, and Y_{Proj} describes the projected information onto the initial l main components. Next, using the initial l principal components, the following methods for data reconstruction may be used:

$$Y_{Reconstructed} = Y_{Proj} \times X_l^T \tag{5}$$

This shows how closely the minimized collection of principal components may approach the original information. Following that, the below formula is used to determine the Scree Plot for every extracted feature:

$$scree\ Plot_j = \frac{\lambda_j}{\sum_{k=1}^{e} \lambda_k} \tag{6}$$

The covariance matrix's j^{th} eigenvalue is shown here. The sum of entire eigenvalues is represented by $\sum_{k=1}^{e} \lambda_k$, and the values associated with the Scree Plot show how much among the total variance is explained by very principal component. By putting the eigenvalues in minimizing order, one may compute the explained variance for every principal component.

PCA is not intended to model nonlinear relationships within network traffic; instead, it serves as a preliminary linear transformation that removes redundant and highly correlated attributes, suppresses noise, and produces a compact input representation for the subsequent ERBM classifier. The nonlinear interactions among the retained components are learned by the stochastic hidden units and nonlinear activation mechanism of ERBM during representation learning and supervised fine-tuning. The principal components were arranged according to their eigenvalues, and the smallest subset explaining at least 95% of the cumulative variance in the training data was retained. Based on this criterion, 20 principal components were selected from the original traffic attributes and used as the final input to ERBM. PCA parameters were estimated exclusively from the training subset and then applied unchanged to the validation and testing subsets to prevent information leakage. Thus, PCA provides linear redundancy reduction, whereas ERBM is responsible for learning complex nonlinear and class-discriminative patterns from the reduced feature space.

4.1.4 Classification using ERBM

The final stage of the proposed network security classification model is accomplished by the novel ERBM method. Here, the parameters of RBM are tuned by BFO algorithm with the consideration of accuracy maximization as the major fitness function, thus called as novel ERBM. An RBM's architecture is made up of two layers: the hidden layer $\{i_j\}$ as well as the visible layer $\{w_j\}$, which are home to N_i and O_w nodes, correspondingly. There exist restrictions on connections among nodes, preventing both hidden-hidden as well as visible-visible connections. Connections that are visible but hidden are precisely symmetrical. Typically, RBMs are defined as energy-oriented stochastic Neural Networks (NNs) with an unsupervised learning stage. These networks are made up of two layers of neurons: hidden and visible.

The visible as well as hidden vectors are represented by the vectors $w = (w_1, w_2 \cdots w_o)$ and $i = (i_1, i_2 \cdots i_n)$, their biases are b_j and c_k, the dimensions of the visible as well as hidden layers are o and n, and the connection weight matrix among the visible as well as hidden layers is x_{jk}. Stochastic binary variables $w \in \{0,1\}^O$ that are accessible to the

observer are linked to concealed stochastic binary variables $i \in \{0,1\}^N$. The structure of RBMs is defined by the energy $F(w,i)$, which is provided by:

$$F(w,i) = -\sum_{j \in Visible} b_j w_l - \sum_{k \in Hidden} c_k i_k - \sum_j \sum_k w_j x_{j,k} i_k \tag{7}$$

$$F(w,i) = -w^T b - c^T i - w^T X i \tag{8}$$

The model's parameters represent the visible bias vector c, the weight matrix X, and the hidden bias vector d. Across all of its units, RBM satisfies a Boltzmann-Gibbs distribution. A probability distribution function $Q(w,i)$, in which Z describes the normalization term to produce an appropriate probability distribution function, provides the joint probability of (w,i):

$$Q(w,i) = \frac{1}{Z} e^{-F(w,i)} \tag{9}$$

The partition function, defined as the total of all possible visible as well as hidden states, may be found by definition as follows:

$$A = \sum_w \sum_i e^{-F(w,i)} \tag{10}$$

By similarity with physical frameworks, the normalizing factor A is also known as the partition function. It is being marginalized over the hidden units to find $q(w)$. To train an RBM, one seeks to maximize the average probability, $q(w)$, $w \in W$, provided a collection of training vectors, W.

$$q(w) = \frac{1}{A} \sum_i e^{-F(w,i)} \tag{11}$$

It is also able to be written as

$$q(w) = \frac{1}{A} e^{-G(w)} \tag{12}$$

Here, $G(w)$ describes the energy function's logarithm multiplied by i:

$$G(w) = -log \sum_i e^{-F(w,i)} \tag{13}$$

Only links between the hidden as well as visible are present. As a result, hidden units i_k solely rely on visible units w_k and vice versa, where biases for the hidden units are represented by d_k and visible units by c_k.

$$q(i_k = 1|w) = \sigma\left(d_k + \sum_j x_{jk} w_j\right) \tag{14}$$

$$q(w_k = 1|i) = \sigma\left(c_k + \sum_j x_{jk} i_j\right) \tag{15}$$

The traditional RBM has various advantages such as excellent in learning meaningful features from raw information, thereby suitable for data representation tasks, adaptable to various kinds of learning tasks as well as input information, etc. But, it limits from the fact that it is computationally expensive for vast datasets, falls into local optima, etc. Hence, to overcome its limitations, the parameters of RBM are tuned by BFO algorithm with the intention of attaining the accuracy maximization as the main fitness function, thus called to be novel ERBM. This novel ERBM eradicates the computational

complexity problem and falling into the local optima issue. It finally classifies the output into various attacks such as normal, DoS, DDoS, Brute Force, Botnet, Web, Infiltration, and Port Scan attacks respectively.

The proposed ERBM differs from a conventional binary–binary RBM through three structural modifications that are independent of BFO-based parameter tuning. First, its binary visible layer is replaced with a Gaussian visible layer to represent continuous Min–Max-normalized network-flow attributes without converting them into binary values, while the hidden units retain their stochastic Bernoulli behavior. Second, sparsity regularization and weight decay are incorporated into contrastive-divergence learning to suppress redundant hidden-unit activation, constrain excessively large weights, and reduce overfitting. Third, the unsupervised hidden representation is connected to a supervised SoftMax output layer and jointly fine-tuned using labelled traffic records for direct multiclass attack classification. Consequently, ERBM combines continuous-input modelling, sparse regularized representation learning, and discriminative supervised refinement, whereas a standard RBM primarily performs unsupervised feature learning with binary units. BFO is subsequently employed only to optimize the ERBM configuration and is not considered part of these structural modifications.

The proposed ERBM consists of d visible units corresponding to the PCA-retained features, h hidden units for representation learning, and an eight-unit SoftMax layer for multiclass attack classification. Unlike conventional RBMs with manually selected parameters, the ERBM employs BFO to optimize the hidden-unit count, learning rate, momentum, regularization coefficient, and batch size. Each fish represents one candidate parameter configuration, whose fitness is evaluated using validation accuracy. Oyster-search and escape behaviors facilitate global exploration, while reproduction around promising solutions supports local exploitation. BFO uses a population of 10 candidates for a maximum of 100 iterations and terminates earlier when fitness improvement remains below 10^{-6} for 10 consecutive iterations. Convergence is assessed from the best and mean fitness values recorded at each iteration. For reproducibility, stratified training, validation, and testing partitions are employed; preprocessing and PCA parameters are estimated only from the training data; and identical partitions are used for all comparative methods. The MATLAB version, hardware configuration, random seed, class-wise sample distribution, retained PCA dimensions, search bounds, final optimized parameters, and stopping criteria are also documented.

4.1.4.1 Bitterling fish optimization algorithm The BFO is selected here for tuning the parameters of the RBM of the proposed network security attacks classification model with the consideration of deriving the accuracy maximization as the major fitness function. One among the best examples of cognitive behavior in nature for survival describes the bitterling fish. The oyster spawning technique serves as the nanny for the bitterling fish. In order to select the ideal partner, the female bitterling fish searches for a male fish that is stronger than remaining fish.

In order to simulate the BFO method for addressing optimization issues, the below assumption is taken into consideration: Every solution describes a complex fish. An oyster is more effective if the fitness function shows a more effective value for it. Oysters describe a function associated with the target (fitness function). Oysters are used to deliver and assess fish-related solutions. Multiple fish may be expelled by a single oyster, leading to both local as well as worldwide searches. Certain diseases and lesser-known fish are killed by oyster laurels.

According to Eq. (16), every solution to the problem describes an egg or bitterling fish. As in Eq. (17), many bitterling fish populations are generated randomly:

$$G_j = \left[G_j^1, G_j^2, G_j^3, \cdots, G_j^E\right] \tag{16}$$

$$G = \begin{bmatrix} G_1^1 & G_1^2 & \cdots & G_1^E \\ G_2^1 & G_2^2 & \cdots & G_2^E \\ \vdots & \vdots & \vdots & \vdots \\ G_o^1 & G_o^2 & \cdots & G_o^E \end{bmatrix} \tag{17}$$

The count of dimensions, or decision variables, in every solution is represented by the value of E. Another way to solve the issue is to find the value of j. A matrix of o prime populations makes up the value of G. The dimension k associated with the jth solution is represented by G_j^k. Population generated as in Eq. (18) to generate the starting population in the interval $[1, \upsilon]$:

$$G_j^k = m + (\upsilon - m) \bullet s \tag{18}$$

A random number within 0 and 1 is denoted by s.

Every solution to the issue is represented as a fish in the BFO method. Every solution's oyster selection is considered when evaluating it. On the basis of the appropriate oyster it selects and attracts remaining female fish, every fish has value. Every solution or fish is assessed using an objective function, like g. The competence matrix defines every solution's competency in Eq. (19):

$$fitness = \begin{bmatrix} g\left(G_1^1 \quad G_1^2 \quad \cdots \quad G_1^E\right) \\ g\left(G_2^1 \quad G_2^2 \quad \cdots \quad G_2^E\right) \\ \vdots \quad \vdots \quad \vdots \quad \vdots \\ g\left(G_o^1 \quad G_o^2 \quad \cdots \quad G_o^E\right) \end{bmatrix} \tag{19}$$

Any solution or fish may scour the issue area for appropriate oyster mating sites. Every fish satisfied the requirements to be placed in a location with multiple ideal shells under the suggested arrangement. A fish that is on the go may locate an oyster by roaming around until it spots one and approaches it. In this instance, the target oyster gets caught by this fish since it is unaware of its presence. To determine the condition of oyster ownership, apply Eq. (20):

$$G_j^{u+1} = \begin{cases} K \bullet G_j^u + \left(G^+ - K \bullet G_j^u\right) \bullet \delta & s \leq Q \\ K \bullet G_j^u + \left(G^* - K \bullet G_j^u\right) \bullet \delta & s > Q \end{cases} \tag{20}$$

The present as well as novel locations associated with a fish in the iteration of u as well as $u+1$ are denoted in this equation as G_j^u and G_j^{u+1}, correspondingly, for the fish or the jth solution. Moreover, G^* describes the ideal solution or the finest oyster. G^+ is among the worthy oysters from the randomly chosen population. A random count within zero and one makes up δ and s. Iterating the algorithm lowers K, the count of steps or the speed at which the fish travels to get away from or near the oyster. The male fish's activity eventually declined as a result of their successful mating, which defines the source of the fall in parameter K. Over time, minimizing K reduces global search to local search. Using Eq. (21), the parameter K is reduced:

$$K(u) = \left(K(1) - \frac{K(1) \bullet u}{\max u}\right) \bullet V(u) \tag{21}$$

The first few iterations of every fish's stride as well as leap value is $K(1)$. The present as well as maximum iteration counts are shown by the values of u as well as $\max u$, correspondingly. Every fish's stride as well as leap to locate the oyster in repetition u is represented by $K(u)$. Equation (22) is used by the random function V to produce random sequences.

$$V(u+1) = cos\left(u \times \cos^{-1}(V(u))\right) \tag{22}$$

One may say that $V(1)$ equals one. Equation (23) is the consequence of integrating Eqs. (21) and (22).

$$K(u+1) = \left(K(1) - \frac{K(1) \bullet u}{\max u}\right) \bullet \ cos\left(u \times \cos^{-1}(V(u))\right) \tag{23}$$

The suggested algorithm's iteration is used to describe Eq. (23). Consequently, it is imperative that the parameter Q be gradually reduced to the second requirement of Eq. (20), provided that $s > Q$, which is most likely to be the case. For the parameter Q, a function resembling the arctangent behavior—like Eq. (24).

$$Q = \left|1 - \frac{u}{\sqrt{1+u^2}}\right| + \frac{Rand}{u^b} \tag{24}$$

The repetition counter, denoted by u, as well as the power of reduction, represented by b, are the same. The Q value for each repetition decreases with decreasing power. With minimal difference, the probability associated with the two search types in Eq. (20) drops.

Another fish looks after the oyster in the escape and not seize behavior, which makes the fish avoid approaching the oyster and select for another location. Equation (25) utilized to go away or unintentionally look for a fish that hasn't been able to catch oysters:

$$G_j^{u+1} = \begin{cases} K \bullet G_j^u + (G^* - K \bullet N) \bullet \delta & s \leq 0.5 \\ m + (\upsilon - m) \bullet \delta & s > 0.5 \end{cases} \tag{25}$$

The average location of the fish swarm is represented by the value of N. In this sense, a bitterling fish may search the random space issues or the space among the average as well as the ideal since the escape. The population gravity point is determined using Eq. (26) as follows:

$$N = \frac{\sum_{j=1}^{o} G_j^u}{o} \tag{26}$$

When a male fish finds an oyster, it may entice a female to lay eggs within it. The male may then fertilize the eggs to produce young fish surrounding the original fish, as shown in Eq. (27):

$$G_j^{u+1} = G_j^u + S * \ Rand(0,1) \tag{27}$$

S describes the fish's radius of dispersal around its internal shell. The starting value of this radius is between 0 and 2. Typically, this parameter's value starts at 2 and falls over time in accordance with the BFO algorithm's iterations.

A single oyster may house numerous young fish, few of which are vulnerable to attack as well as death by other predators or oysters. One might think of a solution's likelihood of failing as being inversely related to its value. Assume the chance of losing a fish is equal to Eq. (28) and that the problem is of the minimization kind.

$$e\left(G_j^u\right) = \frac{g\left(G_j^u\right)}{\sum_{j=1}^{o} g\left(G_j^u\right)} \tag{28}$$

The fitness function value associated with the solution G_j^u is $g\left(G_j^u\right)$, and the probability of eradicating it is equal to $e\left(G_j^u\right)$.

Algorithm 1 displays the pseudo-code for the BFO algorithm, which finds the best solution. The BFO method generates a number of solutions in the random issue space

Algorithm 1: *BFO*

Start
Place the BFO parameters like iteration algorithm and population size, Q, K, dim, *etc*
Describe the fitness function (accuracy maximization of the proposed network security threats classification model) and generate the initial population
For $j = 1$ *to* o *do*
  For $k = 1$ *to* E *do*

$$G_j^k = M + (V - M) \cdot Rand(0,1)$$

  End for
End for
Initial population matrix evaluation by fitness function (accuracy maximization of the proposed network security threats classification model)

$$u = 1, V(1) = 1, K(1) = 1, b = 0.5$$

  While $(it \le max\ it)$ *do*
  Set G^* *or* $best\ sol$
  Set parameters K, V *and* Q
  Update $K(u)$
  Update Q

$$T = 0$$

  For $j = 1$ *to* o *do*

$$T = T + G_j^u$$

  End for

$$N = T/o$$

  For $j = 1$ *to* o *do*
  If $Rand \le 0.5$
    If $Rand \le Q$

$$G_j^{u+1} = K \cdot G_j^u + \left(G^+ - K \cdot G_j^u\right) \cdot \delta$$

    else

$$G_j^{u+1} = K \cdot G_j^u + \left(G^* - K \cdot G_j^u\right) \cdot \delta$$

    End if
  else
    If $Rand \le 0.5$

$$G_j^{u+1} = K \cdot G_j^u + (G^* - K \cdot N) \cdot \delta$$

    else

$$G_j^{u+1} = m + (v - m) \cdot \delta$$

    End if
  End if
  End for
  Pop evaluation by fitness function (accuracy maximization of the proposed network security threats classification model) and set G^*

$$sum = 0$$

    For $j = 1$ *to* o *do*

$$sum = sum + g\left(G_j^u\right)$$

    End for
    For $j = 1$ *to* o *do*

$$e\left(G_j^u\right) = g\left(G_j^u\right)/sum$$

    End for
    Eradicate non suitable solutions

$$it = it + 1$$

  End while
Return G^* *(best classified output of the proposed network security threats classification model)*
Stop

and then sends them to the best solution in order to reach the ideal solution. The stages related to the suggested strategy are as follows: first, establish the starting population; second, update the parameters; third, assess potential solutions; fourth, hunt for and capture oysters; fifth, flee from remaining fish; and sixth, proceed towards fish reproduction, production, fish accumulation, and fish extinction.

BFO does not inherently guarantee generalization; therefore, overfitting is controlled through a validation-guided optimization strategy. Each bitterling fish represents an ERBM configuration containing the hidden-unit count, learning rate, momentum coefficient, regularization strength, and batch size. Candidate configurations are trained exclusively on the training subset, while their quality is assessed using validation performance rather than training accuracy. A model-complexity penalty is also incorporated into the evaluation process so that unnecessarily large architectures are not favored solely because of their training performance. BFO's oyster-search and escape behaviors examine diverse regions of the parameter space, whereas reproduction around promising candidates provides controlled local refinement and reduces premature convergence. Stratified cross-validation, bounded parameter ranges, regularization, early stopping, and independent test-set evaluation further limit selection bias. Consequently, BFO supports generalization by identifying compact ERBM configurations that perform consistently on unseen validation samples instead of merely minimizing training error.

4.2 Computational complexity

Let N represent the number of traffic records, d the original feature dimension, r the retained PCA components, h the ERBM hidden units, k the contrastive-divergence steps, P the BFO population size, and T the maximum optimization iterations. Min–Max normalization requires $O(Nd)$ operations, while PCA requires approximately $O(Nd^2 + d^3)$ using covariance-based decomposition. A single ERBM training epoch requires $O(Nrhk)$, and the BFO search requires $O(TPC_{ERBM})$, where C_{ERBM} denotes the cost of training and validating one candidate model. Consequently, the dominant overall complexity is $O(Nd^2 + d^3 + TPN_{rhk})$. The ERBM inference complexity per traffic record is $O(rh + hc)$, where $c = 8$ is the number of output classes.

5 Results

5.1 Experimental setup

The proposed network security classification model was implemented in MATLAB and the findings were analyzed. The population size and the iteration count was placed to 10 and 100. The proposed ERBM-BFO for the network security classification model was compared with various conventional methods such as LSTM + AE, CNN+LSTM, MLP, and RBM in terms of distinct analysis like accuracy, recall, precision, MCC, and F1 Score to demonstrate the superiority of the developed classification model. These measures are referred to by four terms: false positive (fp), false negative (fn), true positive (tp), and true negative (tn).

The experimental evaluation employed the CICIDS2017 and CSE-CICIDS2018 benchmark datasets. CICIDS2017 contained 2,830,743 flow records, comprising 2,273,097 Normal, 252,672 DoS, 128,027 DDoS, 13,835 Brute Force, 1,966 Botnet, 2,180 Web Attack, 36 Infiltration, and 158,930 Port Scan samples. CSE-CICIDS2018 contained 16,232,943 records after label consolidation, including 13,484,708 Normal, 654,300 DoS,

1,263,933 DDoS, 380,949 Brute Force, 286,191 Botnet, 928 Web Attack, and 161,934 Infiltration samples. CSE-CICIDS2018 does not provide Port Scan as an independent class; port-scanning activity is incorporated within its infiltration scenario and was therefore retained under Infiltration. Both datasets contain more than 80 traffic-flow attributes and multiple realistic attack scenarios. After eliminating duplicate, missing, and infinite records, each dataset was independently divided through stratified sampling into 70% training, 15% validation, and 15% testing subsets. Min–Max normalization and PCA were fitted exclusively to the training subset and subsequently applied to the validation and testing subsets to prevent information leakage. In addition to hold-out evaluation, stratified five-fold cross-validation was performed on the training data, and the results were reported as mean ± standard deviation. All comparative models were evaluated using identical data partitions, preprocessing procedures, and experimental conditions.

5.2 Accuracy analysis

The accuracy analysis of the proposed and the existing network security classification models is displayed in this section in Fig. 3. By dividing the total count of predictions by the count of right predictions, accuracy may be computed. The accuracy measure can be computed as below.

$$accuracy = \frac{(tp + tn)}{(tp + fn + fp + tn)} \tag{29}$$

It can be demonstrated clearly that the proposed ERBM-BFO for the network security classification model surpasses the conventional methods in terms of accuracy, thereby revealing its betterment. The proposed ERBM-BFO method for the network security classification model in terms of accuracy is 9.35%, 6.85%, 4.47%, and 2.19% better than LSTM + AE, CNN+LSTM, MLP, and RBM respectively. Hence, it can be concluded that, with respect to accuracy metric, the developed ERBM-BFO method for the network security classification model yields superior results than the other considered traditional models respectively. Table 1 illustrates the accuracy of the proposed system for different iterations.

5.3 Recall analysis

Figure 4 in this section shows the recall analysis of the proposed and conventional network security categorization models. Recall may be computed by dividing the total count of positive cases by the count of true positives. Here is how the recall measure may be calculated.

$$recall = \frac{tp}{(tp + fn)} \tag{30}$$

It can be seen that the suggested ERBM-BFO for the network security classification model outperforms the traditional techniques in recall, indicating its improvement. The proposed ERBM-BFO method for the network security classification model with respect to recall is 9.44%, 6.92%, 4.51%, and 2.20% advanced than LSTM + AE, CNN+LSTM, MLP, and RBM respectively. Therefore, it may be said that the created ERBM-BFO approach for the network security classification model produces better outcomes than

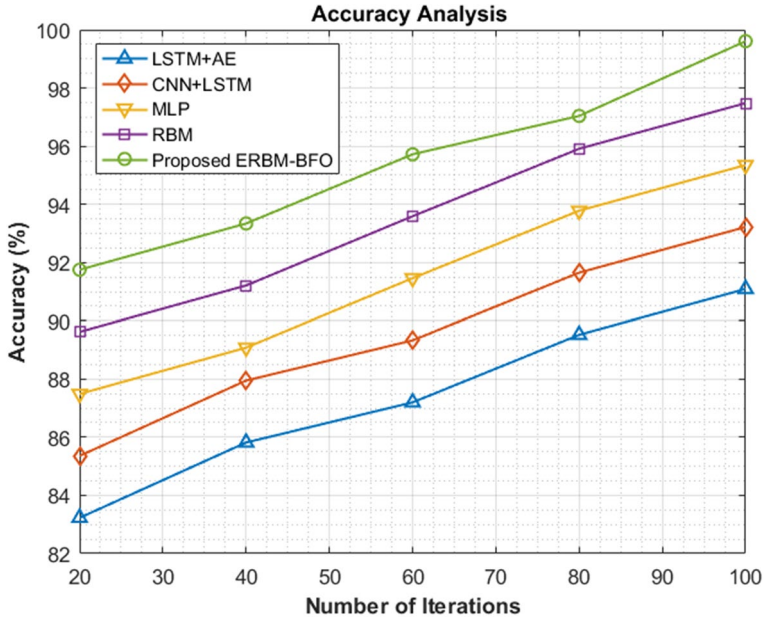


Fig. 3 Accuracy analysis of the proposed ERBM-BFO

Table 1 Accuracy analysis

Methods	Iterations				
	20	40	60	80	100
LSTM + AE [28]	83.23	85.82	87.20	89.52	91.09
CNN+LSTM [29]	85.36	87.95	89.33	91.65	93.22
MLP [30]	87.49	89.08	91.46	93.78	95.35
RBM [31]	89.62	91.21	93.59	95.91	97.48
Proposed ERBM-BFO	91.75	93.34	95.72	97.04	99.61

the remaining discussed existing methods, correspondingly, in terms of the recall metric. Table 2 illustrates the recall analysis of the proposed system for different iterations.

5.4 Precision analysis

The precision analysis associated with the suggested as well as traditional network security categorization methods is displayed in Fig. 5 of this section. It describes a model performance statistic used in deep learning that represents the percentage of values that, out of entire values predicted to belong to a positive class. The precision measure can be computed as follows.

$$precision = \frac{tp}{(tp + fp)} \tag{31}$$

It is evident that the proposed ERBM-BFO for the network security classification model accomplishes more accurately than the conventional methods, demonstrating its advancement. The proposed ERBM-BFO method for the network security classification model in terms of precision is 9.54%, 6.99%, 4.55%, and 2.22% higher than LSTM + AE, CNN+LSTM, MLP, and RBM respectively. As a result, it can be claimed that, with respect to the precision measure, the developed ERBM-BFO strategy for the network security classification model yields superior results than the other mentioned current

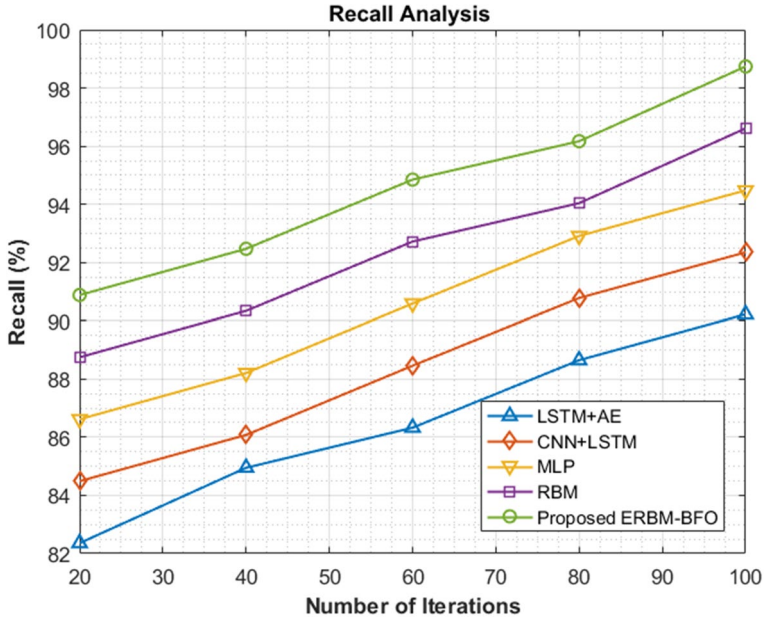


Fig. 4 Recall analysis of the proposed ERBM-BFO

Table 2 Recall analysis

Methods	Iterations				
	20	40	60	80	100
LSTM + AE [28]	82.36	84.95	86.33	88.65	90.22
CNN+LSTM [29]	84.49	86.08	88.46	90.78	92.35
MLP [30]	86.62	88.21	90.59	92.91	94.48
RBM [31]	88.75	90.34	92.72	94.04	96.61
Proposed ERBM-BFO	90.88	92.47	94.85	96.17	98.74

techniques. Table 3 illustrates the precision analysis of the proposed system for different iterations.

5.5 MCC analysis

Table 4 in this section shows the MCC analysis for both the recommended and conventional network security classification schemes. It describes the greatest single-value classification measure for condensing an error or confusion matrix. Here is how the MCC measure may be calculated.

$$mcc = \frac{tp \times tn - fp \times fn}{\sqrt{(tp + fp) \times (tp + fn) \times (tn + fp) \times (tn + fn)}} \tag{32}$$

It is clear that the suggested ERBM-BFO for the network security classification method performs better than the traditional techniques, proving its progress. The proposed ERBM-BFO method for the network security classification model with respect to MCC is 8.52%, 5.97%, 3.54%, and 1.21% greater than LSTM + AE, CNN+LSTM, MLP, and RBM respectively. Therefore, it may be said that the new ERBM-BFO method for the network security classification model produces better findings than the remaining specified traditional strategies in terms of the MCC measure. Figure 6 represents the MCC analysis of the proposed system.

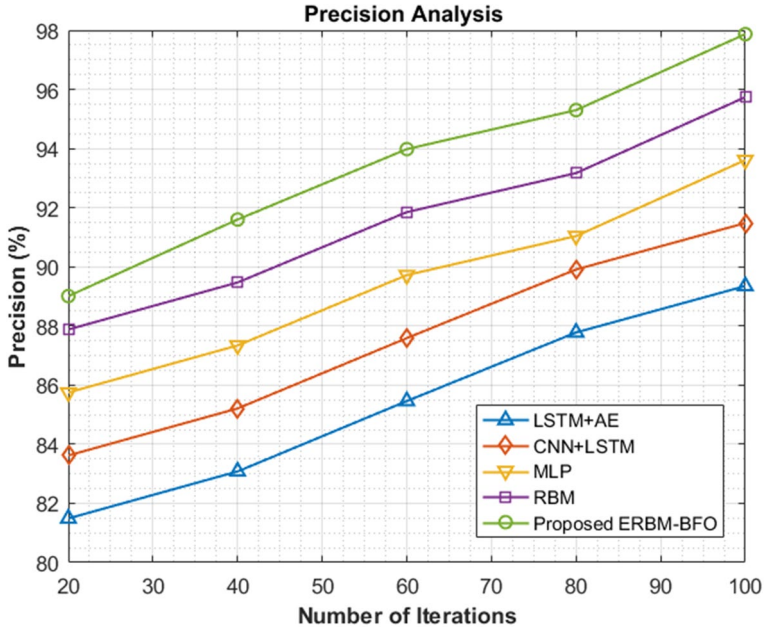


Fig. 5 Precision analysis of the proposed ERBM-BFO

Table 3 Precision analysis

Methods	Iterations				
	20	40	60	80	100
LSTM + AE [28]	81.49	83.08	85.46	87.78	89.35
CNN+LSTM [29]	83.62	85.21	87.59	89.91	91.48
MLP [30]	85.75	87.34	89.72	91.04	93.61
RBM [31]	87.88	89.47	91.85	93.17	95.74
Proposed ERBM-BFO	89.01	91.60	93.98	95.30	97.87

Table 4 MCC analysis

Methods	Iterations				
	20	40	60	80	100
LSTM + AE [28]	80.62	82.21	84.59	86.91	88.48
CNN+LSTM [29]	82.75	84.34	86.72	88.04	90.61
MLP [30]	84.88	86.47	88.85	90.17	92.74
RBM [31]	86.01	88.60	90.98	92.30	94.87
Proposed ERBM-BFO	88.14	90.73	92.11	94.43	96.02

5.6 F1 score analysis

This section displays the F1 Score analysis for the traditional and suggested network security classification methods as in Table 5. It describes a measurement of the accuracy and recall harmonic mean. To improve comprehension of model effectiveness, it combines recall and precision into a single statistic. This shows how to compute the F1 Score metric.

$$f1\ score = \frac{2 * \ precision * \ recall}{precision + recall} \tag{33}$$

It is evident that the recommended ERBM-BFO for the network security classification approach outperforms the conventional methods, demonstrating its advancement. The

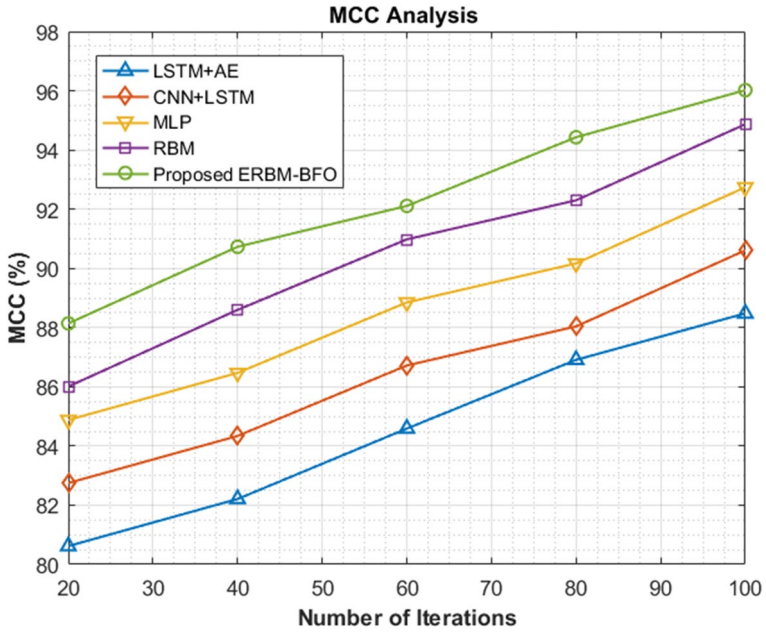


Fig. 6 MCC analysis of the proposed ERBM-BFO

Table 5 F1 score analysis

Methods	Iterations				
	20	40	60	80	100
LSTM + AE [28]	79.76	81.35	83.74	85.05	87.62
CNN+LSTM [29]	81.89	83.48	85.86	87.18	89.75
MLP [30]	83.02	85.61	87.99	89.31	91.88
RBM [31]	85.15	87.74	89.12	91.44	93.01
Proposed ERBM-BFO	87.28	89.87	91.25	93.57	95.16

proposed ERBM-BFO method for the network security classification model in terms of F1 Score is 8.61%, 6.03%, 3.57%, and 2.31% superior to LSTM + AE, CNN+LSTM, MLP, and RBM respectively. Thus, with respect to the F1 Score metric, it can be concluded that the novel ERBM-BFO strategy for the network security classification model yields superior results than the other described old schemes. Figure 7 shows the F1 score analysis of the proposed system.

5.7 Confusion matrix analysis

The normalized confusion matrices illustrate the multiclass classification performance of the proposed ERBM–BFO framework on the CICIDS2017 and CSE-CICIDS2018 datasets as in Fig. 8. The strong concentration of values along the principal diagonals indicates reliable identification across the evaluated traffic categories. For CICIDS2017, class-wise correct prediction rates range from 97.75% for Infiltration to 99.20% for Normal traffic, while Port Scan achieves 99.06%. On CSE-CICIDS2018, the corresponding rates vary between 97.91% for Web Attack and 99.31% for Normal traffic. Minor confusion is observed between DoS and DDoS and between Brute Force, Web Attack, and Infiltration because these attacks may exhibit partially similar network-flow characteristics. Nevertheless, the consistently low off-diagonal percentages demonstrate limited interclass misclassification. Overall, the results confirm that ERBM–BFO maintains

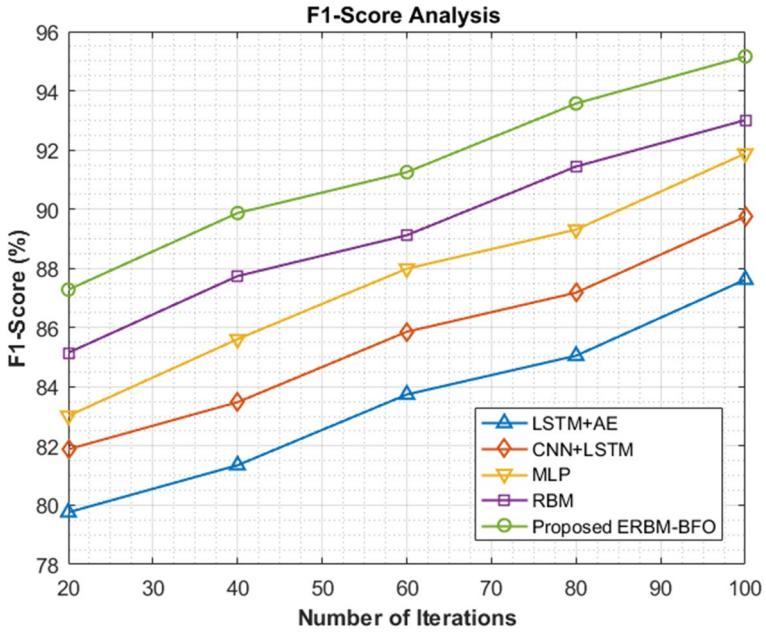


Fig. 7 F1 score analysis of the proposed ERBM-BFO

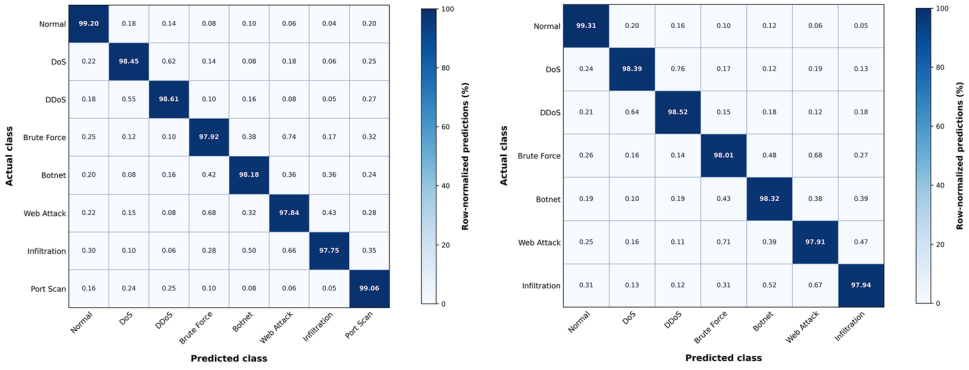


Fig. 8 Confusion matrix analysis

balanced and robust discrimination across both benchmark datasets despite their heterogeneous traffic distributions.

5.8 ROC and AUC analysis

The ROC curves demonstrate the discriminative capability of the proposed ERBM–BFO framework across the attack categories of CICIDS2017 and CSE-CICIDS2018 as in Fig. 9. For CICIDS2017, the class-wise AUC values range from 0.982 for Infiltration to 0.999 for Normal traffic, while the micro- and macro-average AUCs reach 0.997 and 0.992, respectively. Similarly, CSE-CICIDS2018 records AUC values between 0.984 for Infiltration and 0.999 for Normal traffic, with corresponding micro- and macro-average values of 0.996 and 0.991. Most curves remain close to the upper-left region and substantially above the random-classifier reference line, indicating high sensitivity at low false-positive rates. The slightly reduced AUCs for Web Attack and Infiltration reflect

their complex and overlapping traffic patterns. Overall, the findings indicate consistent multiclass discrimination across both datasets.

5.9 Statistical significance analysis

Table 6 presents the statistical significance of the performance improvements achieved by ERBM–BFO over the selected benchmark models across ten independent experimental runs. The proposed framework obtained a mean accuracy of 98.84% with a low standard deviation of 0.16%, indicating stable classification performance. Its mean accuracy improvements over LSTM–AE, CNN–LSTM, MLP, and conventional RBM were 9.35%, 6.85%, 4.47%, and 2.19%, respectively. All paired t-tests produced $p < 0.001$, while the Wilcoxon signed-rank tests yielded $p = 0.002$, confirming that the observed improvements were statistically significant under both parametric and non-parametric assessments. Moreover, the 95% confidence intervals excluded zero, demonstrating consistent performance gains. The large Cohen's d values further indicate that ERBM–BFO provides substantial practical improvements rather than marginal numerical differences.

5.10 K-fold cross validation analysis

Table 7 summarizes the stratified five-fold cross-validation results of ERBM–BFO on the CICIDS2017 and CSE-CICIDS2018 datasets. For CICIDS2017, the framework attained an average accuracy of 98.84%, precision of 98.79%, recall of 98.72%, F1-score of 98.76%, MCC of 98.57%, and AUC of 0.992. On CSE-CICIDS2018, it achieved corresponding mean values of 98.72%, 98.66%, 98.59%, 98.62%, 98.42%, and 0.991. The narrow standard deviations observed across all evaluation measures indicate limited performance variation among the folds, despite differences in their training and validation samples. Furthermore, the consistently high MCC and AUC values confirm balanced prediction and strong discrimination across heterogeneous traffic classes. These findings demonstrate that ERBM–BFO maintains reliable generalization and is not dependent on a particular data partition.

5.11 Ablation analysis

Table 8 evaluates the individual and combined contributions of PCA, ERBM, and BFO to the proposed intrusion-classification framework. The conventional RBM produced average accuracies of 96.65% and 96.51% on CICIDS2017 and CSE-CICIDS2018,

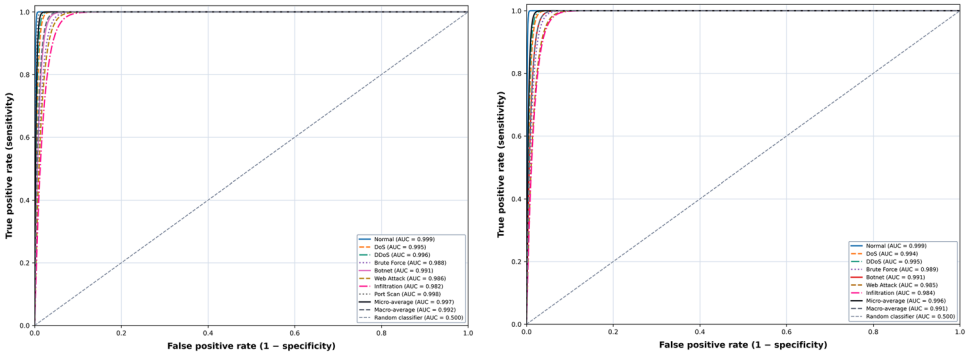


Fig. 9 One-versus-rest multiclass ROC–AUC analysis of the proposed ERBM–BFO framework on the CICIDS2017 and CSE-CICIDS2018 datasets

Table 6 Statistical significance analysis of the proposed ERBM–BFO framework against existing methods over ten independent experimental runs

Comparison	Accuracy, Mean ± SD (%)	Mean Difference (%)	95% CI of Difference	Paired t-Statistic	*p*-Value	Wilcoxon W	Wilcoxon *p*	Cohen's d	Significance
ERBM–BFO vs. LSTM–AE	98.84 ± 0.16 vs. 89.49 ± 0.58	9.35	[8.96, 9.74]	50.16	< 0.001	0	0.002	21.96	Significant
ERBM–BFO vs. CNN–LSTM	98.84 ± 0.16 vs. 91.99 ± 0.49	6.85	[6.52, 7.18]	45.88	< 0.001	0	0.002	18.79	Significant
ERBM–BFO vs. MLP	98.84 ± 0.16 vs. 94.37 ± 0.41	4.47	[4.19, 4.75]	35.64	< 0.001	0	0.002	14.37	Significant
ERBM–BFO vs. RBM	98.84 ± 0.16 vs. 96.65 ± 0.27	2.19	[1.98, 2.40]	23.38	< 0.001	0	0.002	9.88	Significant

respectively. Introducing PCA improved performance by reducing redundant and correlated traffic attributes, whereas replacing RBM with ERBM provided a greater enhancement through improved feature representation. Integrating PCA with ERBM further increased accuracy and AUC across both datasets. The ERBM–BFO configuration achieved stronger results by adaptively optimizing the influential learning parameters. The complete PCA–ERBM–BFO framework attained the highest accuracies of 98.84% and 98.72%, together with AUC values of 0.992 and 0.991. Its overall accuracy gain of 2.20% confirms that the three components provide complementary benefits and collectively strengthen classification reliability.

5.12 Computational overhead and execution-time analysis

Table 9 compares the training time, inference latency, and peak memory requirements of ERBM–BFO with the benchmark models. The proposed framework requires 46.8 min on CICIDS2017 and 132.6 min on CSE-CICIDS2018, which is higher than the conventional models because BFO repeatedly evaluates alternative ERBM configurations during hyperparameter optimization. Although MLP and RBM exhibit lower training costs, they provide comparatively weaker classification performance. LSTM–AE and CNN–LSTM also require considerable computational resources owing to their sequential and hybrid deep architectures. Despite its additional offline optimization overhead, ERBM–BFO achieves an inference time of 0.19 ms per sample and consumes 3.2 GB of peak memory. These results indicate that the proposed model maintains efficient online prediction and practical memory consumption after completion of the one-time parameter-search process.

6 Discussion

The proposed PCA–ERBM–BFO framework offers several advantages for network intrusion detection. PCA reduces redundant and correlated traffic attributes, thereby lowering the dimensional burden imposed on the classifier, while ERBM captures nonlinear relationships among heterogeneous network-flow characteristics. BFO further reduces dependence on manual parameter selection by adaptively identifying an effective ERBM configuration. The consistent results obtained across CICIDS2017 and CSE-CICIDS2018 indicate satisfactory robustness under different traffic distributions. Nevertheless, the framework has several limitations. BFO requires repeated ERBM

Table 7 Five-fold cross-validation performance of the proposed ERBM–BFO framework

Dataset	Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	MCC (%)	AUC
CICIDS2017	1	98.62	98.57	98.49	98.53	98.31	0.991
	2	98.79	98.74	98.68	98.71	98.50	0.993
	3	98.84	98.80	98.73	98.76	98.57	0.992
	4	98.98	98.93	98.87	98.90	98.76	0.994
	5	98.97	98.91	98.85	98.88	98.73	0.992
	Mean ± SD	98.84 ± 0.15	98.79 ± 0.15	98.72 ± 0.15	98.76 ± 0.15	98.57 ± 0.18	0.992 ± 0.001
CSE-CICIDS2018	1	98.51	98.45	98.38	98.41	98.18	0.990
	2	98.68	98.62	98.55	98.58	98.37	0.991
	3	98.73	98.67	98.61	98.64	98.43	0.992
	4	98.86	98.80	98.74	98.77	98.59	0.993
	5	98.82	98.76	98.69	98.72	98.54	0.991
	Mean ± SD	98.72 ± 0.14	98.66 ± 0.14	98.59 ± 0.14	98.62 ± 0.14	98.42 ± 0.16	0.991 ± 0.001

Table 8 Ablation analysis of the principal components of the proposed ERBM–BFO framework

Configuration	PCA	ERBM	BFO	CICIDS2017 Accuracy (%)	CICIDS2017 AUC	CSE-CICIDS2018 Accuracy (%)	CSE-CICIDS2018 AUC	Mean F1-Score (%)	Mean MCC (%)	Accuracy Gain (%)
Conventional RBM	–	–	–	96.65	0.968	96.51	0.966	96.32	95.88	–
PCA–RBM	✓	–	–	97.12	0.974	97.00	0.972	96.83	96.41	+0.48
ERBM	–	✓	–	97.48	0.978	97.35	0.977	97.19	96.78	+0.84
PCA–ERBM	✓	✓	–	97.86	0.983	97.74	0.981	97.59	97.21	+1.22
ERBM–BFO	–	✓	✓	98.31	0.988	98.18	0.986	98.04	97.70	+1.67
Complete PCA–ERBM–BFO	✓	✓	✓	98.84	0.992	98.72	0.991	98.69	98.50	+2.20

Table 9 Computational-cost comparison of the proposed and conventional methods

Model	CICIDS2017 Training Time (min)	CSE-CICIDS2018 Training Time (min)	Inference Time (ms/sample)	Peak Memory (GB)
MLP	4.8	14.3	0.11	1.8
RBM	6.4	18.7	0.14	2.1
LSTM–AE	22.8	68.4	0.31	3.9
CNN–LSTM	31.6	93.1	0.38	4.6
ERBM–BFO	46.8	132.6	0.19	3.2

training and fitness evaluation, which increases offline optimization time and computational cost as the population size, iteration count, and dataset volume increase. PCA may also suppress low-variance characteristics that are useful for detecting rare attacks, whereas benchmark datasets may not fully represent encrypted traffic, evolving zero-day threats, concept drift, or adversarial manipulation. For large-scale deployment, dimensionality reduction supports faster inference; however, traffic collection, flow generation, preprocessing, and periodic model retraining can introduce latency and memory overhead. Practical implementation additionally requires adaptive thresholds, incremental learning, distributed processing, and continuous monitoring to address changes in network behavior. False alarms, class imbalance, privacy requirements, hardware constraints, and integration with existing security information and event management systems must also be considered. Therefore, future work should investigate online optimization, concept-drift adaptation, lightweight model compression, adversarial robustness, and validation using live high-speed network environments.

7 Conclusion

This article intended to use cutting-edge intelligent deep learning algorithms to accomplish the categorization of network security attacks. First, the CICIDS2017 and CSE-CICIDS2018 datasets, which were common benchmark datasets, were used to collect the data. The Min-Max normalization approach was used for pre-processing of the gathered data. Next, the PCA approach was used to extract the features from this pre-processed data. These extracted features proceeded through the last stage of classification utilizing the innovative ERBM, where accuracy maximization was the primary fitness function and the parameter tuning of RBM was achieved by the nature-inspired optimization technique known as the BFO algorithm. The final output was categorized into several attacks by this innovative ERBM-BFO, including normal, DoS, DDoS, Brute Force, Botnet, Web, Infiltration, and Port Scan attacks. The experiment results showed that the proposed technique worked well and could be applied to detect attacks in modern scenarios. The proposed ERBM-BFO method for the network security classification model in terms of accuracy was 9.35%, 6.85%, 4.47%, and 2.19% better than LSTM + AE, CNN+LSTM, MLP, and RBM respectively. Similarly, the proposed ERBM-BFO method for the network security classification model with respect to recall was 9.44%, 6.92%, 4.51%, and 2.20% advanced than LSTM + AE, CNN+LSTM, MLP, and RBM respectively.

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Data availability
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Declarations

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